

E-Commerce Sales & Customer Analytics

1. Executive Summary

This project analyzes e-commerce sales, customer behavior, profitability, and operational performance using an end-to-end analytics workflow.

The analysis was performed on **138,116 records across 46 columns** using **Python, MySQL, and Power BI**. The project involved data cleaning, exploratory data analysis, business-focused SQL analysis, data modeling, DAX measures, and interactive dashboard development.

The analysis identified important patterns in customer value, sales performance, profitability, and fulfillment, providing a data-driven foundation for business monitoring and decision-making.

2. Objectives

- Analyze overall **sales and profitability** performance.
 - Understand **customer behavior, loyalty, and customer value**.
 - Compare performance across **customer segments, regions, sales channels, and other business dimensions**.
 - Evaluate **delivery, returns, cancellations, and fulfillment performance**.
 - Identify meaningful business trends and opportunities through data.
 - Build interactive Power BI dashboards for **business reporting and decision support**.
-

3. Data & Methodology

Data

- **Dataset size:** 138,116 rows × 46 columns.
- Key fields included:
 - Order and customer information
 - Sales, discounts, taxes, shipping costs
 - Net sales and profit
 - Delivery and return information
 - Customer ratings and lifetime value
 - Marketing and payment information

Methodology

Python

- Performed data exploration and quality checks.
- Analyzed missing values, duplicates, categorical and numerical variables.
- Cleaned the data based on business meaning.
- Created features such as:
 - `order_month`
 - `delivery_delay`
 - `discount_status`
- Identified outliers and retained potential genuine business values.
- Conducted exploratory data analysis across sales, customers, profitability, delivery, returns, and marketing.

MySQL

- Loaded the cleaned dataset into MySQL.
- Answered business questions using:
 - Aggregations
 - `GROUP BY`
 - `CASE WHEN`
 - Joins
 - Subqueries
 - Filtering and sorting

Power BI

- Built a relational data model and DateTable.
- Created DAX measures for key KPIs such as:
 1. Total Sales
 2. Total Profit
 3. Total Orders
 4. Total Customers
 5. Average Order Value
 6. Profit Margin
 7. Repeat Customer %
 8. Return Rate
 9. On-Time Delivery %
- Developed three dashboards:
 1. **Sales & Profitability Overview**
 2. **Customer Behavior & Loyalty**
 3. **Operational Performance & Fulfillment**

4. Key Insights

- **Consumer customers generated approximately \$97.20M in net sales**, making them the highest-sales customer segment in the analyzed dataset.
- **Repeat customers generated approximately \$176.70M in net sales and \$75.95M in profit**, substantially exceeding the contribution from new customers.
- New customers contributed approximately **\$435K in net sales and \$193K in profit**, highlighting a large difference in contribution between repeat and new customers within this dataset.
- Delivery performance varied across the analyzed periods, providing an opportunity to monitor periods with higher delivery delays.
- Regional and shipping-method comparisons showed that some differences were relatively small, demonstrating the importance of focusing dashboards on metrics that provide meaningful business separation.
- The analysis emphasized **data-model validation**: product/category sales visuals were avoided where the available table relationships could not reliably support the calculation.

Business Value

The project demonstrates an end-to-end approach to transforming raw e-commerce data into **cleaned datasets, business analysis, actionable insights, and interactive reporting**. It also highlights the importance of validating data quality and relationships before using analytical results for business decisions.